Reinforcement Learning for Query Rewriting and Schema Selection

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Project Overview

The project develops a single reinforcement learning agent for two connected decisions in Text-to-Cypher: whether and how to rewrite a natural-language question, and which graph-schema triples to select. Both tasks are modeled inside one Gymnasium environment. The agent first operates in a rewrite phase, then moves to a schema-selection phase, and finally receives end-to-end feedback. The main objective is to learn a compact decision policy that improves schema prediction without performing unnecessary rewrites.

The dataset contains 484 Hetionet questions paired with Cypher queries and gold schema triples. It is divided into 339 training, 73 validation, and 72 test examples using a hop-stratified split. The retriever indexes only training examples, and validation is used for checkpoint selection. Gold schema information is used for demonstrations and rewards during training, but is never included in the agent observation at evaluation time.

Reinforcement Learning Formulation

Environment and episode

One episode contains two phases. In the REWRITE phase, the agent can keep the original question, choose append or keywords mode, add entity or relation tokens, and execute STOP_REWRITE. This stop action changes the phase but does not terminate the episode. In the SCHEMA phase, the same policy sequentially selects schema triples and eventually executes STOP_SCHEMA, which terminates the episode. The environment uses action masking so that only actions valid for the current phase and state receive nonzero probability.

Component	Definition
State	122 values: original and rewritten question features, 24 question-to-triple compatibility values, rewrite-token mask, selected-schema mask, phase, mode, and progress.
Action space	63 discrete actions: 35 rewrite tokens, two rewrite modes, STOP_REWRITE, 24 schema triples, and STOP_SCHEMA.
Transition	Rewrite actions modify the current query. STOP_REWRITE recomputes schema compatibility and enters schema selection. Schema actions modify the selected triple set.
Termination	Only STOP_SCHEMA terminates the episode.

Reward and credit assignment

The main RL challenge is delayed credit assignment: a rewriting decision is useful only if it improves retrieval or helps the later schema-selection decisions. The final design therefore provides phase-specific feedback. At STOP_REWRITE, the environment compares retrieval with the rewritten query against retrieval with the original

question. Keeping the original question has zero counterfactual reward; useful rewrites receive positive feedback; ineffective or harmful rewrites pay action costs and a no-gain penalty.

rewrite reward = retrieval score after rewriting - retrieval score before rewriting - action costs

During schema selection, each action receives the change in schema F1 minus a small complexity cost. The terminal reward combines final schema F1, exact schema completion, and a penalty for missing triples. This reward structure prevents the much larger schema reward from completely hiding the contribution of earlier rewrite actions.

Agent and Training

The agent is a neural Actor-Critic with one shared two-layer encoder, two phase-conditioned actor heads, and one critic. The rewrite head and schema head operate over different action subsets, but they belong to one policy because they share the state representation, trajectory, optimizer, return, and value function. The model has 40,768 trainable parameters. Masked categorical distributions are used to sample valid actions.

Training starts with behavioral cloning from training-only pseudo-oracle trajectories. The pseudo-oracle compares keep, append, and keywords rewrites using retrieval utility, then demonstrates the gold schema actions. Rewrite trajectories are oversampled because useful rewriting is a minority decision. After 40 epochs, action accuracy reaches 0.935.

Actor-Critic fine-tuning runs for 10,000 episodes. The loss contains the policy-gradient objective, critic mean-squared error, entropy regularization, and a small rewrite-imitation loss that decays from 0.10 to 0.01. The critic reduces variance by estimating state value, entropy preserves exploration, gradient clipping improves stability, and validation checkpointing prevents the final model from being selected from training reward alone.

Results

Method	Schema F1	Exact schema	Precision	Recall	Ret exact	Ret recall	Rewrite tokens
Behavioral cloning	0.865	0.694	0.912	0.845	0.583	0.850	1.042
Unified Actor-Critic	0.920	0.792	0.958	0.907	0.583	0.863	0.167
Unified AC without rewriting	0.916	0.778	0.951	0.907	0.569	0.850	0.000

Actor-Critic fine-tuning improves schema F1 from 0.865 to 0.920 and exact schema accuracy from 0.694 to 0.792 compared with behavioral cloning. The learned policy also reduces mean rewrite length from 1.042 to 0.167 tokens. This indicates that RL improves both the schema decisions and the stopping policy.

The no-rewrite ablation is the critical comparison. It uses the same trained model but forces STOP_REWRITE immediately. Allowing learned rewriting increases schema F1 from 0.916 to 0.920, exact schema accuracy from 0.778 to 0.792, retrieval exact hit from 0.569 to 0.583, and retrieval recall from 0.850 to 0.863. The gain is modest, corresponding to approximately one additional exact prediction, but every quality metric moves in the expected direction.

The final policy keeps 68 of 72 test questions unchanged, uses keywords mode for two, and append mode for two. This is not a collapse of the rewriting component. It is the learned solution to a cost-sensitive stopping problem: most original questions are already suitable, while a small number benefit from intervention.

Interpretation and Conclusion

The project demonstrates several central RL concepts: sequential state transitions, heterogeneous actions, delayed credit assignment, learned stopping, reward shaping, action masking, exploration, value-based variance reduction, imitation initialization, and policy ablation. An earlier reward formulation caused the agent to eliminate rewriting because downstream schema reward dominated the weak retrieval signal. Counterfactual phase reward and question-to-triple compatibility features repaired this failure by making rewriting both observable and directly accountable.

The main conclusion is therefore not that rewriting should be performed frequently. The result is that one Actor-Critic policy can learn when rewriting has positive expected downstream value and otherwise stop, before selecting a compact graph schema. On the held-out test set, the unified agent reaches 0.920 schema F1 and 0.792 exact schema accuracy. The remaining limitations are the small test set, only two three-hop test examples, a lightweight TF-IDF retriever, constrained rather than generative rewrites, and evaluation of schema prediction rather than final Cypher execution.